20 pt Prove that if the linear programming has solutions, then at least one such solution is a basic optimal point.

0 pt Consider the subproblem in trust-region method,

$$\min_{\substack{p\in\mathbb{R}^n \\ \|p\|\le\Delta}} f + g^\top p + \frac{1}{2}p^\top Bp.$$

p^* is a global solution of the subproblem if and only if p^* is feasible and there is a scalar $\lambda \ge 0$ such that the following conditions are satisfied,

$$(B+\lambda I)p^* = -g,$$

$$\lambda(\Delta - \|p^*\|) = 0,$$

$(B+\lambda I)$ is positive semidefinite.

[Hint: Using KKT conditions and Problem 1]

1.11　试卷: 数值代数与优化

原网页

1.12　26 春期末考

本次考试两大题除最后一问外完全开卷, 可自由使用 ai, 需在 3 小时内完成并上传代码. 两道大题的最后一问是另发试卷, 在最后一小时内完成, 完全开放任何纸质资料. 总时长 4 小时.

数据集下载: `https://pan.baidu.com/s/1Twv-LoniVcjVI1fAYztqew?pwd=antc`

一、 Given a sparse matrix $A \in \mathbb{R}^{m\times n}$ and a vector $b \in \mathbb{R}^m$. We look for a vector x that minimizes the norm

$$\|Ax - b\|_2.$$

The matrix A and the vector b are given in the dataset.

(1) (10 pt) Find a property, e.g., sparsity, low-rank, etc., of the given matrices and vectors in the dataset. Based on this property, propose three potential methods to solve the problem.
[Mention the method names if they were discussed in class, or propose a short pseudocode and explain why the methods are suitable.]

(2) (15 pt) Design an algorithm to address the problem with matrices and vectors in the given dataset.

(3) (5 pt) Discuss the requirements of A and b such that your designed algorithm can be applied.

(4) (10 pt) Write the value of x_3 for each matrix-vector pair in the dataset.
[Keep 6 decimal digits in your answer.]

(5) (10 pt) (This problem was answered under closed-book conditions.) Given matrix $L \in \mathbb{R}^{p\times n}$, and a parameter $\lambda > 0$, consider the regularized least-squares problem

$$\min \|Ax - b\|_2^2 + \lambda\|Lx\|_2^2$$

Give a necessary and sufficient condition such that the above minimization problem has a unique minimizer.

二、 Consider the following constrained optimization problem:

$$\min_{\substack{\mathrm{ReLU}(c_i^\top x+\beta_i)=0\\ i=K+1,\dots,M}} \sum_{i=1}^{K} w_i \,\mathrm{ReLU}(c_i^\top x+\beta_i).$$

In the above problem, the ReLU function is defined as

$$\mathrm{ReLU}(z)=\max(0,z).$$

(1) (10 pt) Write a KKT-like condition for the above system. More precisely, please write the necessary conditions for the optimality of the above optimization problem.
[If the above problem cannot be answered directly, you could replace ReLU by ReLU^2 and then write down the KKT conditions.]

(2) (10 pt) Design and implement an algorithm to find a feasible point for the optimization problem with given data.
[The algorithm could adopt the property of the given dataset and be adapted to your following optimization solver.]

(3) (10 pt) Design and implement an algorithm to solve the optimization problem with given data.

(4) (10 pt) Apply the designed algorithm to the given matrices. Write the minimum objective values.
[Keep 6 decimal digits in your answer.]

(5) (10 pt) (This problem was answered under closed-book conditions.) Solve this problem by hand

$$\min_{\substack{\mathrm{ReLU}(x_1+2x_2-x_3+4)=0\\ \mathrm{ReLU}(-2x_1+x_2+x_3+3)=0\\ \mathrm{ReLU}(x_1-x_2+3x_3+-6)=0\\ \mathrm{ReLU}(-x_1+4x_2+x_3+7)=0}} \mathrm{ReLU}(2x_1-3x_2-4x_3-2)$$

1.13　25 春期末考

第一题, 除矩阵向量运算外, 任何作为解答的算法需要自行实现, 不能直接调用矩阵分解, 线性方程组求解等库函数. 例如, 若将 LU 分解作为解答, 需要自行实现 LU 分解算法, 不得直接调用 MATLAB 自带 [L,U]=lu(A) 函数.

第二题, 允许调用矩阵分解等库函数, 但不允许直接调用优化求解器.

一、 Given a symmetric matrix $A\in\mathbb{R}^{n\times n}$. We look for a unitary matrix W such that,

$$W=\begin{pmatrix} A & * \\ * & * \end{pmatrix},$$

where $*$ denotes an unspecified block of the matrix. The matrix given in the dataset is A^2.

(1) (10 pt) Give an if-and-only-if condition on A such that the unitary matrix W exists.

(2) (10 pt) Assume A could be embedded into a unitary matrix W, where the size of W is a power of 2, i.e., $W\in\mathbb{R}^{2^k\times 2^k}$ for some integer k. Design a unitary matrix W.

(3) (20 pt) Design an algorithm taking A^2 as the input and compute the application of W on a given vector.
[Your algorithm should include factorization of A. The algorithm could be tailored to the specific structure of A^2 in the dataset.]

(4) (10 pt) Let n be the size of A. Embed A into W of size $2n$. Write the value of

$$\begin{pmatrix} 1 & 1 & 0 & \cdots & 0 \end{pmatrix} W(e_1 + e_{n+1}),$$

for each A^2 in the dataset, where $e_i \in \mathbb{R}^{2n}$ is a zero column vector with 1 at the i-th position.

二、 Given two nonsingular matrices $A \in \mathbb{R}^{m\times k}$ and $B \in \mathbb{R}^{k\times n}$ with $k < \min(m, n)$, a weight vector $c \in \mathbb{R}^n$, and a vector $b \in \mathbb{R}^m$. Consider the following linear program,

$$\min_{\substack{ABx=b \\ x\geq 0}} c^\top x.$$

(1) (10 pt) Write the KKT condition for above system.

(2) (20 pt) Design and implement an algorithm to solve the linear program.
[The algorithm could be tailored to the specific structure of matrices in the dataset.]

(3) (5 pt) Based on the given datasets, analyze the computational complexity, and give a complexity upper bound for your designed algorithm.

(4) (15 pt) Apply the designed algorithm to the given matrices. Write the minimum objective values.
[Keep 6 decimal digits in your answer.]